

Additive, Geoadditive and Structured Additive Models

[Chapter 43](#) learned to estimate one unknown curve. Real data sets have a dozen covariates, and one unknown surface in a dozen dimensions is hopeless: no neighbourhood of a point holds enough observations to average. The way out is a structural assumption weaker than linearity but far stronger than nothing — the covariates act **additively**, each through its own unknown function — which turns a hopeless problem into a stack of one-dimensional ones held together by a single penalized criterion.

This chapter builds that framework and keeps widening it. A coefficient may vary smoothly with a second covariate. A term may be a spatial effect, penalized towards its neighbours on a map or towards a smooth surface over coordinates. A term may be a random intercept, a penalized term whose penalty is the identity. All of these are *one* model — a structured additive predictor — with *one* estimation theory: a quadratic penalty per term, effective degrees of freedom from a trace, smoothing parameters from a restricted likelihood, and a Bayesian reading in which every penalty is a prior. The last section puts a link and a response distribution in front of that predictor and closes the circle with [Part VIII](#).

What you need. [Chapter 6](#) (projections, the trace as a rank, Proposition 6.3.4, and Theorem 6.6.2), [Chapter 26](#) (collinearity and the variance inflation factor, Definition 26.2.1), [Chapter 27](#) and [Chapter 30](#) (ridge, penalized least squares and effective degrees of freedom, Theorem 27.2.2 and Definition 30.1.1), [Chapter 29](#) (cross-validation and information criteria, Theorem 29.3.2, Definition 29.3.2, Proposition 29.5.1), [Chapter 32](#) (mixed models, BLUP and REML, Definition 32.2.1, Theorem 32.3.1, Theorem 32.4.2), [Chapter 34](#) (the generalized linear model and IRLS, Definition 34.2.1, Theorem 34.4.1), [Chapter 38](#) (quasi-likelihood and dispersion, Definition 38.2.1, Proposition 38.3.2), [Chapter 39](#) (the Bayesian machinery, Definition 39.4.1, Proposition 39.5.1) and, above all, [Chapter 43](#), whose P-spline blocks and smoothing-parameter criteria are used here as given. The algebra behind backfitting and the effective degrees of freedom is Part I's: the cyclic trace, the generalized Rayleigh quotient and the characterizations of a positive definite matrix (Theorem 1.6.1, Corollary 1.7.8, Theorem 1.7.4, [Chapter 1](#)) and the covariance of a linear transformation (Theorem 2.2.1, [Chapter 2](#)); [Section 44.4](#) also uses best linear unbiased prediction (Theorem 14.2.8, [Chapter 14](#)).

Remark (Notation in this chapter). A **term** is a pair: a design block $\mathbf{Z}_j$ of size $n \times d_j$ and a nonnegative definite penalty $\mathbf{P}_j$, with coefficients $\boldsymbol{\gamma}_j$ and smoothing parameter λ_j — the $\mathbf{P}$, $\boldsymbol{\gamma}$ and λ of [Chapter 43](#), now carrying a term index. So $\mathbf{Z}$ keeps its meaning from [Part VII](#), the design of the coefficients that are shrunk, and [Section 44.5](#) shows the two uses are the same. Unpenalized columns stay in $\mathbf{X}$ with coefficients $\boldsymbol{\beta}$; $\mathbf{A} = \mathbf{Z}^\top \mathbf{W} \mathbf{Z} + \mathbf{P}$ is the penalized cross-product, $\mathbf{W}$ the working weights of [Chapter 34](#), $\mathbf{S}$ the smoother matrix and $\mathbf{C}_j$ the operator producing the j th component.

44.-1.1 Roadmap

- [The additive model](#)
- [Backfitting and concurvity](#)
- [Varying-coefficient models](#)
- [Spatial effects](#)

- Structured additive regression
- Generalized additive models
- Summary and notes